# CUSTOMER CHURN PREDICTION

Machine Learning Project Report

## Project objective

Develop a customer churn prediction prototype that analyses customer attributes, compares machine-learning models, estimates churn risk, and supports retention planning.

## Project scope

Data preparation • Exploratory analysis • Feature engineering • Model comparison • Threshold and cost analysis • Batch prediction • Business insights • Drift-monitoring prototype

| Project dataset summary | Value |
|---|---|
| Customer records after cleaning | 440,832 |
| Churned customers | 249,999 (56.71%) |
| Non-churned customers | 190,833 (43.29%) |
| Selected prediction pipeline | Random Forest |

Prepared as a prototype using the supplied customer churn dataset. Student name, institution, internship provider and submission date can be added before submission.

# 1. Methodology and Implementation

The source dataset contained customer demographics, tenure, usage frequency, support calls, payment delay, subscription type, contract length, total spend, last interaction and a churn label. An empty row was removed; the cleaned dataset contained 440,832 records with no remaining missing values or duplicate customer IDs.

## Feature engineering

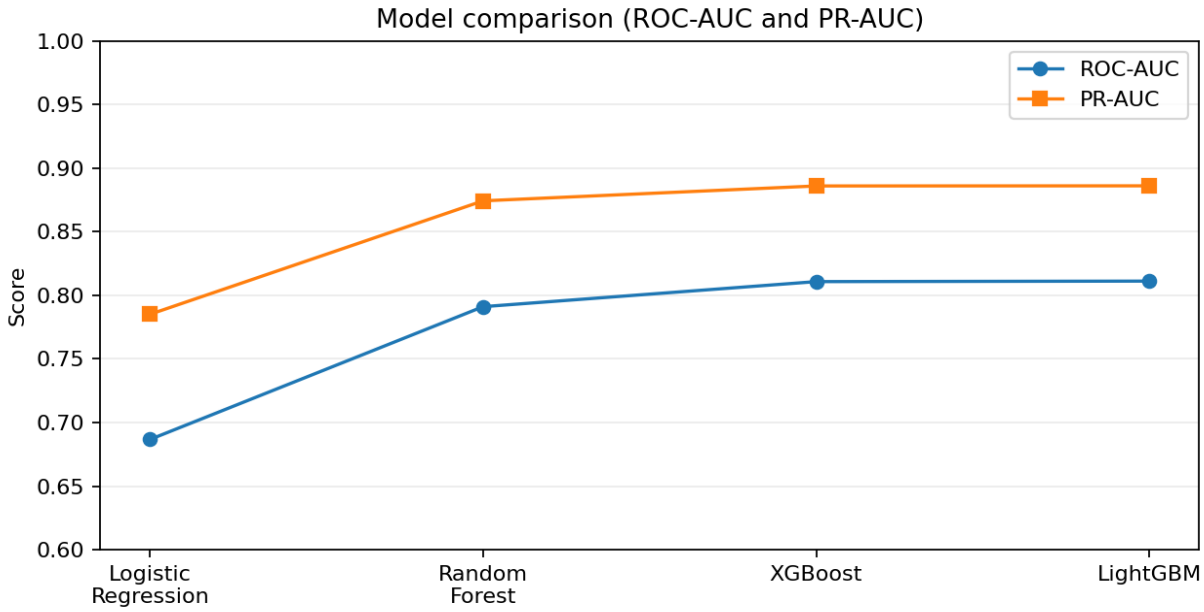
Engineered variables included Usage_Intensity (usage frequency divided by tenure) and Inactivity_Flag (last interaction of at least 20 days). Additional derived variables were explored, but suspicious features based on support calls, payment delay, total spend and contract length were excluded from the strict model feature set.

## Final strict feature set

Age, Tenure, Usage Frequency, Last Interaction, Usage_Intensity, Inactivity_Flag, Gender_Male, Subscription Type_Premium and Subscription Type_Standard.

## Models compared

| Model | Accuracy | ROC-AUC | PR-AUC |
|---|---|---|---|
| Logistic Regression | 61.93% | 0.6867 | 0.7851 |
| Random Forest | 71.78% | 0.7911 | 0.8743 |
| XGBoost | 74.92% | 0.8108 | 0.8860 |
| LightGBM | 74.90% | 0.8112 | 0.8861 |




XGBoost and LightGBM performed very similarly. Random Forest was retained for the current saved prediction pipeline and threshold/cost demonstration; this does not mean it was the top model on every comparison metric.

# 2. Evaluation and Business Findings

The saved Random Forest pipeline uses a provisional probability threshold of 0.30. On the held-out batch prediction file, churn precision was 0.64 and churn recall was 0.86; overall accuracy at this threshold was approximately 64%. The ROC-AUC was 0.7911.

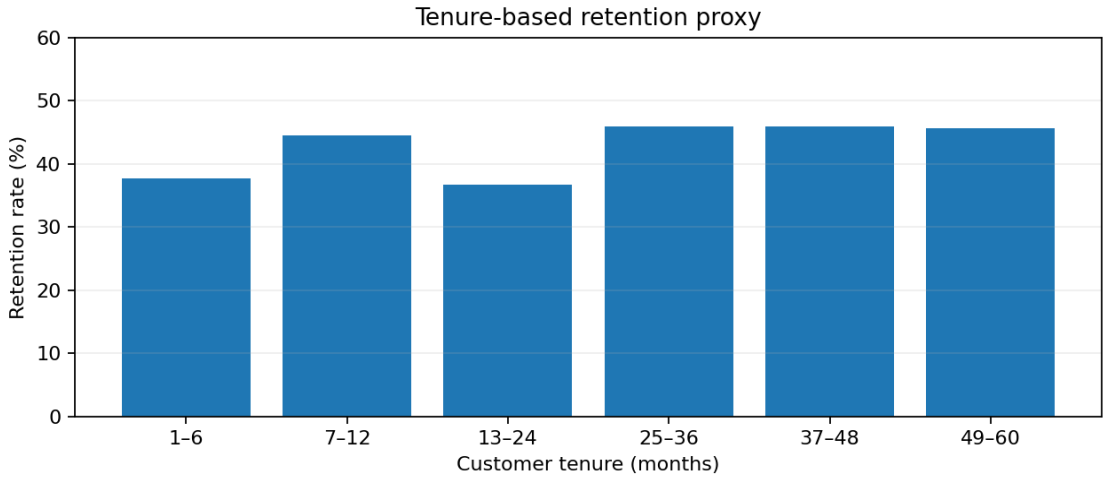
## Threshold and illustrative cost analysis

Thresholds from 0.30 to 0.70 were compared. Under an illustrative cost assumption of 1 unit for a false positive and 5 units for a false negative, threshold 0.30 had the lowest estimated total cost (60,229 units) among the tested thresholds. These are not real monetary costs.

## Business patterns observed

| Segment | Observed churn rate |
|---|---|
| Monthly contracts | 100.00% |
| Payment delay 21–30 days | 100.00% |
| Support calls 5–10 | 99.08% |
| Last interaction 16–30 days | 66.50% |

## Tenure-based retention proxy

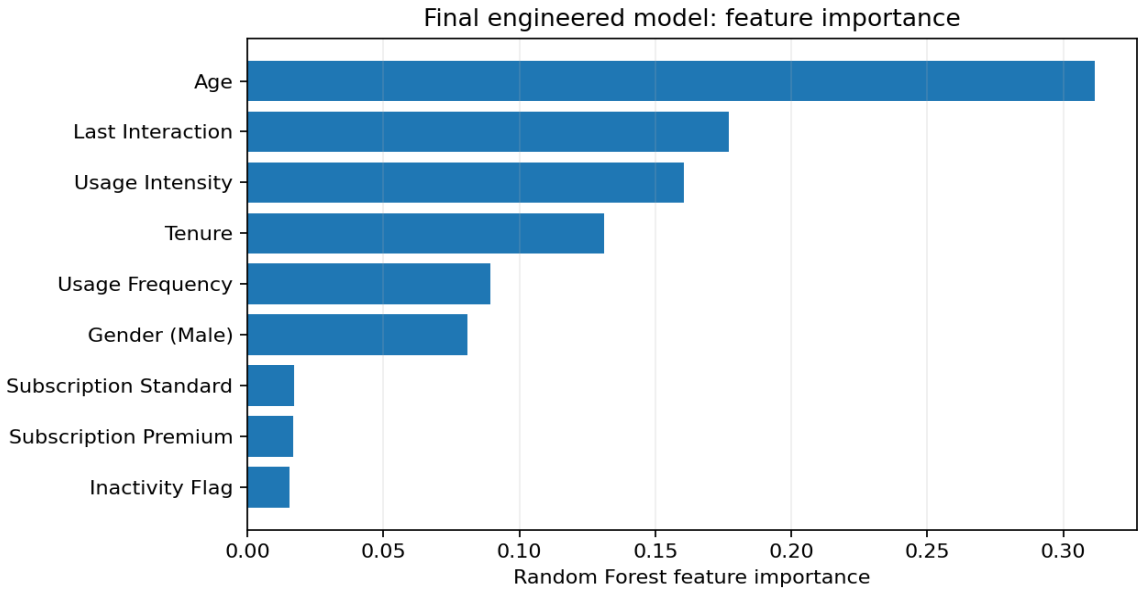



These are cross-sectional tenure-group retention proportions, not longitudinal cohort retention curves. The dataset does not contain verified signup dates or repeated customer observations.

# 3. Explainability, Deployment and Limitations

## Explainability

Feature importance from the final engineered Random Forest ranked Age, Last Interaction, Usage_Intensity and Tenure among the strongest features. Approximate TreeSHAP explanations were generated for a 20-row sample from the final model and saved as summary and bar plots. The small sample and approximate calculation should be disclosed when interpreting the plots.




## Prototype deployment components

The project saves a trained model bundle, generates batch predictions, and scores a small synthetic new-customer file. A drift-monitoring script compares a small incoming sample with the reference data. The five-row sample is only a demonstration and is not sufficient for reliable drift conclusions.

## Limitations and future work

The dataset is cross-sectional and has no verified event timestamps, signup dates or censoring information; therefore genuine time-based cohorts and Kaplan–Meier survival analysis cannot be validated. Perfect-churn patterns in some segments may reflect dataset-generation rules; their cause is unknown. Airflow scheduling, MLflow tracking, CRM/reverse-ETL integration, real-time monitoring and automated retraining are proposed future integrations, not implemented services. Model results require validation on independent, representative business data.

## Conclusion

The project delivers an end-to-end churn prediction prototype covering data preparation, model comparison, risk scoring, explainability, business analysis and preliminary monitoring. Its results are promising within this dataset but should not be interpreted as guaranteed real-world predictive performance.

Dataset source: MatanKriel, “Matan-Kriel-Customer-Churn-dataset”, Hugging Face.
https://huggingface.co/datasets/MatanKriel/Matan-Kriel-Customer-Churn-dataset

# Deployment Model Clarification

This note accompanies the original Customer Churn Prediction project report to distinguish the model used for the original model-comparison and analysis work from the compact model currently used by the Streamlit application.

## Compact deployment Random Forest

The live application uses a separately trained compact Random Forest model. Its held-out test metrics are shown below. The classification threshold is 0.30.

| Metric | Compact deployment model |
|---|---|
| Accuracy | 74.97% |
| ROC-AUC | 0.8085 |
| PR-AUC | 0.8849 |
| Precision at threshold 0.30 | 59.41% |
| Recall at threshold 0.30 | 95.07% |

## How to read the original report

The four-model comparison table in the original report describes the earlier model-comparison experiments. The original report's threshold/cost results and its 0.7911 ROC-AUC refer to the earlier Random Forest pipeline, not the compact deployment model listed above. They should not be treated as metrics for the live Streamlit application.

## Explainability note

The existing approximate TreeSHAP plots were generated from a 20-row sample of the earlier final engineered Random Forest. They are retained as an explainability analysis of that model and are not explanations of the compact deployment model.

## Scope and limitations

The compact model's metrics are test-set results on the available dataset. They do not establish performance on independent real-world customer data. The dataset and prototype limitations described in the original report still apply.

For submission, keep this clarification page immediately after the original report.